**PAST EXAMINATION QUESTIONS (FP2)**

**TOPIC 8: MATRICES (2) – SYSTEM OF EQUATIONS**

**1.** (a) Show that the system of equations

$$x - 2y - 4z = 1,$$
$$x - 2y + kz = 1,$$
$$-x + 2y + 2z = 1,$$

where $k$ is a constant, does not have a unique solution. **[2]**

(b) Given that $k = -4$, show that the system of equations in part (a) is consistent. Interpret this situation geometrically. **(There is a line of intersection with other plane.)** **[3]**

(c) Given instead that $k = -2$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically. **(Two parallel plane, not identical)** **[2]**

(d) For the case where $k \neq 2$ and $k \neq 4$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically. **(The three planes form a prism)** **[2]**

**Oct/Nov 2020/21/23/3**

**2.** It is given that $a$ is a positive constant.

(a) Show that the system of equations

$$ax + (2a + 5)y + (a + 1)z = 1,$$
$$-4y = 2,$$
$$3y - z = 3,$$

has a unique solution and interpret this situation geometrically. **(Three planes intersect at a single point.)** **[3]**

**Oct/Nov 2020/22/9 (part)**

**3.** (a) Find the values of $a$ for which the system of equations

$$3x + y + z = 0,$$
$$ax + 6y - z = 0,$$
$$ay - 2z = 0,$$

does not have a unique solution. ($\boldsymbol{a = 4, -9}$) **[3]**

**May/June 2020/21/22/8 (part)**

**4.** (a) (i) Find the set of values of $a$ for which the system of equations

$$x - 2y - 2z + 7 = 0,$$
$$2x + (a - 9)y - 10z + 11 = 0,$$
$$3x - 6y + 2az + 29 = 0,$$

has a unique solution. ($\boldsymbol{a \neq 5, a \neq -3}$) **[4]**

(ii) Given that $a = -3$, show that the system of equations in part (i) has no solution. Interpret this situation geometrically. **(Two parallel planes with one non-parallel plane)** **[3]**

**Specimen Paper 2020/2/8 (part)**

**5.** Find the two values of the constant $k$ for which the equation

$$kx + y + z = 2,$$
$$x + ky + z = -1,$$
$$x + y + kz = -1,$$

have no unique solution. ($\boldsymbol{k = 1, -2}$) **[4]**

Show that, for one of these values of $k$, the equations have no solution, and solve the equations for the other value of $k$. ($\boldsymbol{k = 1}$, **no solution**; $\boldsymbol{k = -2, x = t - 1, y = z = t}$) **[3]**

**May/June 2016/13/3**

**6.** Find the value of $a$ for which he system of equations

$$x - y + 2z = 4,$$
$$x + ay - 3z = b,$$
$$x - y + 7z = 13,$$

where $a$ and $b$ are constants, has no unique solution. ($\boldsymbol{a = -1}$) **[3]**

Taking $a$ as the value just found,

(i) find the general solution in the case $b = -5$, ($\boldsymbol{x = 0.4 + t, y = t, z = 1.8}$) **[4]**

(ii) interpret the situation geometrically in the case $b \neq -5$. **(Planes form a prism)** **[1]**

**Oct/Nov 2014/11/12/13/5**

**7.** Find the set of values of $a$ for which the system of equations

$$ax + y + 2z = 0,$$
$$3x - 2y = 4,$$
$$3x - 4y - 6az = 14,$$

has a unique solution. $\left(\boldsymbol{a} \in \mathbb{R}, \boldsymbol{a} \neq \frac{\mathbf{1}}{\mathbf{2}}, \mathbf{-2}\right)$ **[4]**

**Oct/Nov 2012/13/2**

**8.** Find the set of values of $a$ for which the system of equations

$$x - 2y - 2z + 7 = 0,$$
$$2x + (a - 9)y - 10z + 11 = 0,$$
$$3x - 6y + 2az + 29 = 0,$$

has a unique solution. ($\boldsymbol{a \neq 5, a \neq -3}$) **[4]**

Show that the system has no solution in the case $a = -3$. **[2]**

Given that $a = 5$,

(i) show that the number of solutions is infinite, **[2]**

(ii) find the solution for which $x + y + z = 2$. $\left(\boldsymbol{x = -1, y = \frac{7}{2}, z = -\frac{1}{2}}\right)$ **[3]**

**May/June 2012/13/10**

**9.** Find the set of values of $a$ for which the system of equations

$$x + 4y + 12z = 5,$$
$$2x + ay + 12z = a - 1,$$
$$3x + 12y + 2az = 10,$$

has a unique solution. ($\boldsymbol{a \in \mathbb{R}, a \neq 8, 18}$) **[4]**

Show that the system does not have any solution in the case $a = 18$. **[2]**

Given that $a = 8$, show that the number of solutions is infinite and find the solution for which $x + y + z = 1$. $\left(\boldsymbol{x = \frac{1}{3}, y = \frac{5}{12}, z = \frac{1}{4}}\right)$ **[5]**

**May/June 2010/11/12/10**

**10.** Show that if $a \neq 3$ then the system of equations

$$2x + 3y + 4z = -5,$$
$$4x + 5y - z = 5a + 15,$$
$$6x + 8y + az = b - 2a + 21.$$

has a unique solution. **[3]**

Given that $a = 3$, find the value of $b$ for which the equations are consistent. ($\boldsymbol{b = 10}$) **[3]**

**Oct/Nov 2006/1/5**